## EXERCISE-I (Conceptual Questions) Build Up Your Understanding

### DEVELOPMENT OF PERIODIC TABLE

**1.** Mendeleev's periodic table is based on :-
(1) Atomic number
(2) Increasing order of number of protons
(3) Electronic configuration
(4) None of the above

**PT0001**

**2.** Which of the following is/are Dobereiners triad :-
(a) P, As, Sb (b) Cu, Ag, Au
(c) Fe, Co, Ni (d) S, Se, Te
Correct answer is :-
(1) a and b (2) b and c (3) a and d (4) All

**PT0002**

**3.** Which of the following sets of elements follows Newland's octave rule :-
(1) Be, Mg, Ca (2) Na, K, Rb
(3) F, Cl, Br (4) B, Al, Ga

**PT0003**

**4.** Which are correct match :-
(a) Eka silicon – Be
(b) Eka aluminium – Ga
(c) Eka mangenese – Tc
(d) Eka scandium – B
(1) b, c (2) a, b, d (3) a, d (4) All

**PT0004**

**5.** Atomic wt. of P is 31 and Sb is 120. What will be the atomic wt. of As, as per Dobereiners triad rule :-
(1) 151 (2) 75.5
(3) 89.5 (4) Unpredictable

**PT0005**

**6.** The places that were left empty by Mendeleev's were, for:-
(1) Aluminium & Silicon
(2) Galium and germinium
(3) Arsenic and antimony
(4) Molybdenum and tungstun

**PT0006**

**7.** Which is not anomalous pair of elements in the Medeleev's periodic table:-
(1) Ar and K (2) Co and Ni
(3) Te and I (4) Al and Si

**PT0007**

**8.** The law of triads is applicable to :-
(1) Os, Ir, Pt (2) Ca, Sr, Ba
(3) Fe, Co, Ni (4) Ru, Rh, Pt

**PT0008**

**9.** Elements which occupied position in the lother meyer curve, on the peaks, were :–
(1) Alkali metals
(2) Highly electro positive elements
(3) Elements having large atomic volume
(4) All

**PT0009**

**10.** In a period the elements are arranged in :–
(1) Decreasing order of nuclear charge
(2) Decreasing order of No. of electrons
(3) Increasing order of nuclear charge
(4) In order of same nuclear charge

**PT0010**

**11.** Which of the following statement is wrong :-
(1) $2^{nd}$ period contain 8 elements
(2) $3^{rd}$ period contains 18 elements
(3) $1^{st}$ period contains two non metals
(4) In p–block, metal, nonmetal and metalloids are present

**PT0011**

**12.** IUPAC name of the element placed just after actinide series :-
(1) Unniltrium (2) Unnilpentium
(3) Unnilquadium (4) Ununbium

**PT0013**

**13.** Which statement is wrong for the long form of periodic table :-
(1) Number of periods are 7 and groups 18
(2) No. of valence shell electrons in a period are same
(3) $III^{rd}$ B group contains 32 elements
(4) Lanthanides and actinides are placed in same group

**PT0014**

**14.** Which of the following statement is false :-
(1) Elements of $ns^2np^6$ electronic configuration lies in $1^{st}$ to $6^{th}$ period
(2) Typical elements lies in $3^{rd}$ period
(3) The seventh period will accommodate thirty two elements
(4) Boron and silicon are diagonally related

**PT0017**

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**15.** Among the Lanthanides the one obtained by synthetic method is :–
(1) Lu (2) Pm (3) Pr (4) Ce
**PT0018**

**16.** Atomic weight of Cl is 35.5 and I is 127. What will be the atomic weight of Br, as per Dobereiners triad rule :-
(1) 81.2 (2) 85 (3) 95 (4) 162
**PT0237**

**17.** Elements which occupied ascending positions on Lothar meyer curve are :-
(1) Alkali metals (2) Alkaline earth metals
(3) Halogens (4) Metaloids
**PT0238**

## PERIOD, GROUP AND BLOCK

**18.** Which of the following set of elements belongs to same period :-
(1) Zn, Cd, Hg (2) Fr, Ra, U
(3) K, Ca, Ag (4) None
**PT0019**

**19.** The element with atomic number Z = 115 will be placed in :-
(1) $7^{th}$ period, IA group (2) $8^{th}$ period, IVA group
(3) $7^{th}$ period, VA group (4) $6^{th}$ period, VB group
**PT0020**

**20.** In 6th period of the modern periodic table, electronic energy levels are in the order :-
(1) 6s, 4f, 5d, 6p (2) 6s, 6p, 4f, 5d
(3) 4f, 5d, 6s, 6p (4) None
**PT0022**

**21.** The IUPAC name of the element which is placed after $Db_{105}$ is the periodic table, will be :-
(1) Un nil pentium (2) Un un nilium
(3) Un nil hexium (4) Un nil quadium
**PT0024**

**22.** The element with atomic number Z=118 will be :–
(1) Noble gas (2) Transition metal
(3) Alkali metal (4) Alkaline earth metal
**PT0026**

**23.** The atom having the valence shell electronic configuration $4s^2\ 4p^2$ would be in:-
(1) Group II A and period 3
(2) Group II B and period 4
(3) Group IV A and period 4
(4) Group IV A and period 3
**PT0027**

**24.** The electronic configuration of d-block elements is exhibited by :-
(1) $ns^{0-2}(n-1)d^{1-10}$ (2) $ns^2\ (n-1)\ d^{10}$
(3) $(n-1)d^{10}s^2$ (4) $ns^2np^5$
**PT0028**

**25.** The element having electronic configuration $4f^{14}\ 5d^0\ 6s^2$ belongs to :-
(1) d–block, $12^{th}$ group
(2) f–block, III B group
(3) f–block, $14^{th}$ group
(4) s–block, $2^{nd}$ group
**PT0030**

**26.** Element with the electronic configuration given below, belong to which group in the periodic table $1s^2,\ 2s^22p^6,\ 3s^23p^63d^{10},\ 4s^24p^64d^{10},\ 5s^25p^3$
(1) $3^{rd}$ (2) $5^{th}$
(3) $15^{th}$ (4) $17^{th}$
**PT0031**

**27.** Which of the following electronic configuration belongs to inert gas elements :-
(1) $ns^2\ (n-1)d^{10}$ (2) $ns^2\ (n-1)s^2p^6$
(3) $ns^2\ np^6$ (4) None
**PT0033**

**28.** From atomic number 58 to 71, elements are placed in ;-
(1) $5^{th}$ period and III A group
(2) $6^{th}$ period and III B group
(3) separate period and group
(4) $7^{th}$ period and IV B group
**PT0034**

**29.** Elements having $ns^2\ np^6$ valence shell electronic configuration lies in :-
(1) '0' gp. and $1^{st}$–$7^{th}$ period
(2) $18^{th}$ gp. and $2^{nd}$–$6^{th}$ period
(3) $18^{th}$ gp and $1^{st}$–$6^{th}$ period
(4) All are correct
**PT0036**

**30.** The electronic configuration of elements X and Z are $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ and $1s^2\ 2s^2\ 2p^5$ respectively. What is the position of element X with respect to position of Z in the periodic table -
(1) Just below Z (2) Just above Z
(3) Left to the Z (4) right to the Z
**PT0038**

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**31.** Which of the following sequence contains atomic number of only representative elements

(1) 55, 12, 18, 53

(2) 13, 33, 54, 83

(3) 3, 33, 53, 87

(4) 22, 33, 55, 66

**PT0039**

**32.** Uranium (At No. - 92) is the last natural element in the periodic table. The last element of the periodic table which is recently discovered is Uub. What will be the total number of transuranic elements in the periodic table :-

(1) 21 (2) 20 (3) 11 (4) 12

**PT0040**

**33.** Which two elements are in same period as well as same group of modern periodic table :-

(1) Z = 23, Z = 31

(2) Z = 65, Z = 66

(3) Z = 52, Z = 87

(4) Z = 58, Z = 46

**PT0041**

**34.** Which of the following statement is not correct for given electronic configuration

$1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^6 4d^{10} 4f^{14}, 5s^2 5p^6\ 5d^{10}, 6s^2$

(1) It belongs to IIB group and $6^{th}$ period

(2) It is liquid at room temperature

(3) It is a transition element

(4) It is not used in high temperature thermometer

**PT0042**

**35.** For the element Z = 120, in which family would you place -

(1) Group 18, Inert gas

(2) Group 2, Alkaline earth metal

(3) Group 15, Nitrogen

(4) Group 3, Inner transition element

**PT0239**

**36.** Which of the following properties is a periodic function -

(1) Heat of fusion

(2) Heat of vaporisation

(3) Energy of atomization

(4) Atomic radius

**PT0240**

**37.** Select the correct match –

$P = ns^2np^4$ (A) Group no = 17, p-block

$Q = (n-1)d^2\ ns^2$ (B) Group no = 4, d-block

$R = (n-2)f^7(n-1)d^1ns^2$ (C) Group no = 16, p-block

$S = (n-1)d^{10}ns^2np^5$ (D) Group no = 3, f-block

(1) (P–A),(Q–B), (R–C), (S–D)

(2) (P–C), (Q–B), (R–D), (S–A)

(3) (P–A),(Q–C), (R–B), (S–D)

(4) (P–B),(Q–D), (R–C), (S–A)

**PT0241**

**38.** No. of letters used for giving symbol to new discovered element which have not been given official name by IUPAC

(1) 4 (2) 3

(3) 2 (4) 1

**PT0242**

**39.** The filling of 4f orbitals begins with

(1) Cerium (2) Promethium

(3) Lanthanum (4) Gadolinium

**PT0243**

**40.** In which of the following blocks, metals are found.

(1) s (2) p

(3) d (4) All

**PT0244**

## Zeff, SCREENING CONSTANT & ATOMIC RADIUS

**41.** The formula for effective nuclear charge is (if σ is screening constant)

(1) $Z - \sigma$ (2) $Z + \sigma$

(3) $Z\,\sigma^{-1}$ (4) $Z\,\sigma$

**PT0046**

**42.** According to Slater rule, Effective nuclear charge in group generally :–

(1) Increases down the group

(2) Decreases down the group

(3) Remains constant

(4) First increases then decreases

**PT0047**

**43.** In sodium atom the screening is due to :-

(1) $3s^2$, $3p^6$ (2) $2s^1$

(3) $1s^2$, $2s^2$, $2p^6$ (4) $1s^2$, $2s^2$

**PT0048**

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**44**. The correct order of size would be:-
(1) Ni < Pd ≃ Pt (2) Pd < Pt < Ni
(3) Pt > Ni > Pd (4) Pd > Pt > Ni
**PT0050**

**45**. Which of the following order of radii is correct
(1) Li < Be < Mg (2) $H^+ < Li^+ < H^-$
(3) O < F < Ne (4) $Na^+ > F^- > O^{-2}$
**PT0051**

**46.** $K^+$, Ar, $Ca^{2+}$ and $S^{2-}$ contains -
(1) Same electronic configuration and atomic volume
(2) Different electronic configuration but same IP.
(3) Same electronic configuration but different atomic volume
(4) None
**PT0052**

**47.** Which of the following is not isoelectronic series :-
(1) $Cl^-$, $P^{3-}$, Ar (2) $N^{3-}$, Ne, $Mg^{+2}$
(3) $B^{+3}$, He, $Li^+$ (4) $N^{3-}$, $S^{2-}$, $Cl^-$
**PT0053**

**48**. Atomic radii of Fluorine and Neon in Angstrom units are given by :-
(1) 0.72, 1.60 (2) 1.60, 1.60
(3) 0.72, 0.72 (4) None of these
**PT0055**

**49**. Which of the following has largest radius :-
(1) $1s^2\ 2s^2\ 2p^6\ 3s^2$
(2) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$
(3) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$
(4) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$
**PT0056**

**50.** Which of the following order of atomic/ionic radius is not correct :–
(1) $I^- > I > I^+$ (2) $Mg^{+2} > Na^+ > F^-$
(3) $P^{+5} < P^{+3}$ (4) Li > Be > B
**PT0057**

**51.** In the lithium atom screening effect of valence shell electron is caused by-
(1) Electrons of K and L shell
(2) Electrons of K shell
(3) Two electrons of 1st and one of 2nd shell
(4) One $e^-$ of K shell
**PT0058**

**52**. Correct order of ionic radii is
(1) $Ti^{4+} < Mn^{7+}$ (2) $^{37}Cl^- < {}^{35}Cl^-$
(3) $K^+ > Cl^-$ (4) $P^{3+} > P^{5+}$
**PT0059**

**53**. $S^{-2}$ is not isoelectronic with :-
(1) Ar (2) $Cl^-$ (3) $HS^-$ (4) $Ti^{+3}$
**PT0061**

**54**. The best reason to account for the general tendency of atomic diameters to decrease as the atomic numbers increase within a period of the periodic table is the fact that
(1) Outer electrons repel inner electrons
(2) Closer packing among the nuclear particles is achieved
(3) The number of neutrons increases
(4) The increasing effective nuclear charge exerts a greater attractive force on the electrons
**PT0062**

**55.** In an anion :-
(1) Number of proton decreases
(2) Protons are more than electrons
(3) Effective nuclear charge is more
(4) Radius is larger than neutral atom
**PT0063**

**56**. Which of the following ion has largest size :-
(1) $F^-$ (2) $Al^{+3}$ (3) $Cs^+$ (4) $O^{-2}$
**PT0065**

**57.** Spot the incorrect order of atomic radii :-
(1) $r_{Cu > Zn}$ (2) $r_{Cl > F}$
(3) $r_{P > S}$ (4) $r_{Sc > Ti}$
**PT0067**

**58**. Which of the following orders of atomic radii are correct :-
(a) Li < Be < Na (b) Ni < Cu < Zn
(c) Ti > V > Cr (d) Ti > Zr ≃ Hf
Correct answer is :-
(1) All (2) a, b
(3) b, c (4) b, d
**PT0068**

**59.** Decreasing order of size of ions is :-
(1) $Br^- > S^{-2} > Cl^- > N^{-3}$
(2) $N^{3-} > S^{-2} > Cl^- > Br^-$
(3) $Br^- > Cl^- > S^{-2} > N^{-3}$
(4) $N^{-3} > Cl^- > S^{-2} > Br^-$
**PT0070**

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**60**. Which of the following statement is wrong
(1) According to Slater, $Z_{eff}$ in group remains constant
(2) In a period atomic size decreases
(3) Screening effect in a period remains constant
(4) In a period atomic radius of inert gas element is maximum

**PT0071**

**61.** Correct order of $r_{metalic}$ for 3d series is ?
(1) Fe < Co < Ni (2) Cr < Mn > Fe
(3) Sc < Ti < V (4) Cu > Zn

**PT0245**

**62.** According to Slater's rule, order of effective nuclear charge for last electron in case of Li, Na and K :-
(1) Li > Na > K (2) K > Na > Li
(3) Na > Li > K (4) Li < Na = K

**PT0076**

**63.** Finding size of an atom (isolated) is a lot more complicated due to :-
(1) Isolation of single atom is complicated
(2) the electron cloud surrounding the atom does not have a sharp boundary.
(3) Both
(4) Isolated atom does not exist

**PT0246**

**64.** The incorrect match is :

| **Element** | **Standard Radius** |
|---|---|
| (A) Metal | Covalent radii |
| (B) Non metal | Covalent radii |
| (C) Metal | Metallic radius |
| (D) Noble gases | van der Waal radii |

(1) A (2) B (3) C (4) D

**PT0247**

**65.** Radii of Noble gases should be compared with which radii of the other elements :
(1) covalent radii (2) metallic radii
(3) van der Waal radii (4) ionic radii

**PT0248**

**66.** Estimation of ionic radii can be done by measuring :-
(1) their covalent radius
(2) metallic radius
(3) distances between cations and anions in ionic crystals.
(4) radius of neutral atom

**PT0249**

## IONISATION POTENTIAL

**67**. Correct order of $I^{st}$ I.P. are :-
(a) Li < B < Be < C
(b) O < N < F
(c) Be < N < Ne
(1) a, b (2) b, c (3) a, c (4) a, b, c

**PT0078**

**68**. The ionisation potential of isotopes of an element will be :-
(1) Same
(2) Different
(3) Depends on atomic masses
(4) Depends on number of neutrons

**PT0079**

**69**. The second ionisation potentials in electron volts of oxygen and fluorine atoms are respectively given by :-
(1) 35.1, 38.3 (2) 38.3, 38.3
(3) 38.3, 35.1 (4) 35.1, 35.1

**PT0080**

**70**. A sudden large jump between the values of $2^{nd}$ and $3^{rd}$ IP of an element would be associated with the electronic configuration :-
(1) $1s^2, 2s^2\ 2p^6, 3s^1$
(2) $1s^2, 2s^2\ 2p^6, 3s^2\ 3p^5$
(3) $1s^2, 2s^2\ 2p^6, 3s^2\ 3p^2$
(4) $1s^2, 2s^2\ 2p^6\ 3s^2$

**PT0081**

**71.** Compared to the first ionisation potential, the value of second ionisation potential of an element is :-
(1) Negligible (2) Smaller
(3) Greater (4) Double

**PT0082**

**72.** In which of the following pairs, the ionisation energy of the first species is less than that of the second :-
(1) $O^-, O^{2-}$ (2) S, P
(3) N, P (4) $Be^+$, Be

**PT0083**

**73.** The correct order of stability of $Al^+$, $Al^{+2}$, $Al^{+3}$ is :-
(1) $Al^{+3} > Al^{+2} > Al^+$
(2) $Al^{+2} > Al^{+3} > Al^+$
(3) $Al^{+2} < Al^+ > Al^{+3}$
(4) $Al^{+3} > Al^+ > Al^{+2}$

**PT0084**

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**74.** Least ionisation potential will be of :-
(1) $Be^{3+}$ (2) H (3) $Li^{+2}$ (4) $He^{+}$
**PT0085**

**75.** Ionisation energy increases in the order :-
(1) Be, B, C, N (2) B, Be, C, N
(3) C, N, Be, B (4) N, C, Be, B
**PT0086**

**76.** Mg forms Mg(II) because of :-
(1) The oxidation state of Mg is + 2
(2) Difference between $I.P_1$ and $I.P_2$ is greater than 16.0 eV
(3) There is only one electron in the outermost energy level of Mg
(4) Difference between $I.P_1$ and $I.P_2$ is less than 11 eV
**PT0087**

**77.** Minimum first ionisation energy is shown by which electronic configuration:-
(1) $1s^2\ 2s^2\ 2p^5$ (2) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$
(3) $1s^2\ 2s^2\ 2p^6\ 3s^1$ (4) $1s^2\ 2s^2\ 2p^6$
**PT0088**

**78.** With reference to ionisation potential which one of the following set is correct :-
(1) Li > K > B (2) B > Li > K
(3) Cs > Li > K (4) Cs < Li < K
**PT0089**

**79.** Successive ionisation energies of an element 'X' are given below (in K. Cal)

| $IP_1$ | $IP_2$ | $IP_3$ | $IP_4$ |
|---|---|---|---|
| 165 | 195 | 556 | 595 |

Electronic configuration of the element 'X' is:-
(1) $1s^2, 2s^2 2p^6, 3s^2 3p^2$ (2) $1s^2, 2s^1$
(3) $1s^2, 2s^2 2p^2$ (4) $1s^2, 2s^2 2p^6, 3s^2$
**PT0090**

**80.** Second IP of which of the element is maximum–
(1) Lithium (2) Oxygen
(3) Nitrogen (4) Fluorine
**PT0091**

**81.** The energy needed to remove one electron from unipositive ion is abbreviated as :-
(1) 1st I.P. (2) 3rd I.P.
(3) 2nd I.P. (4) 1st E.A.
**PT0092**

**82.** Among the following elements (Whose electronic configuration is given below) the one having the highest ionisation energy is
(1) [Ne] $3s^2\ 3p^3$ (2) [Ne] $3s^2\ 3p^4$
(3) [Ne] $3s^2 3p^5$ (4) [Ar] $3d^{10} 4s^2 4p^2$
**PT0093**

**83.** The correct order of decreasing first ionisation energy is :-
(1) Si > Al > Mg > Na (2) Si > Mg > Al > Na
(3) Al > Si > Mg > Na (4) Mg > Li > Al > Si
**PT0094**

**84.** Out of $Na^+$, $Mg^{+2}$, $O^{-2}$ and $N^{-3}$, the pair of species showing minimum and maximum IP would be.
(1) $Na^+$, $Mg^{+2}$ (2) $Mg^{+2}$, $N^{-3}$
(3) $N^{-3}$, $Mg^{+2}$ (4) $O^{-2}$, $N^{-3}$
**PT0095**

**85.** Lowest IP will be shown by the element having the configuration :–
(1) [He] $2s^2$ (2) $1s^2$
(3) [He] $2s^2\ 2p^2$ (4) [He] $2s^2\ 2p^5$
**PT0097**

**86.** Which ionisation potential (IP) in the following equations involves the greatest amount of energy:-
(1) $K^+ \rightarrow K^{+2} + e^-$ (2) $Li^+ \rightarrow Li^{+2} + e^-$
(3) $Fe \rightarrow Fe^+ + e^-$ (4) $Ca^+ \rightarrow Ca^{+2} + e^-$
**PT0099**

**87.** (a) $M^-_{(g)} \rightarrow M_{(g)}$ (b) $M_{(g)} \rightarrow M^+_{(g)}$
(c) $M^+_{(g)} \rightarrow M^{+2}_{(g)}$ (d) $M^{+2}_{(g)} \rightarrow M^{+3}_{(g)}$
Minimum and maximum I.P. would be of :-
(1) a, d (2) b, c
(3) c, d (4) d, a
**PT0101**

**88.** Triad - I [$N^{3-}$, $O^{-2}$, $Na^+$ ]
Triad - II [ $N^+$, $C^+$, $O^+$ ]
Choose the species of lowest IP from triad–I and highest IP from triad–II respectively
(1) $N^{3-}$, $O^+$ (2) $Na^+$, $C^+$
(3) $N^{3-}$, $N^+$ (4) $O^-$, $C^+$
**PT0103**

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**89.** The correct values of ionisation energies (in kJ $mol^{-1}$) of Be, Ne, He and N respectively are

(1) 786, 1012, 999, 1256

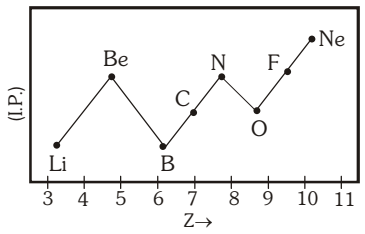
(2) 1012, 786, 999, 1256

(3) 786, 1012, 1256, 999

(4) 786, 999, 1012, 1256

**PT0104**

**90.** Following graph shows variation of I.P. with atomic number in second period (Li – Ne). Value of I.P. of Na (11) will be :-




(1) Above Ne
(2) Below Ne but above O
(3) Below Li
(4) Between N and O

**PT0105**

**91.** Consider the following ionisation reactions

$A(g) \longrightarrow A^{+}(g) + e^{-}$ IE in (KJ/mol) is $A_1$

$A^{+}(g) \longrightarrow A^{+2}(g) + e^{-}$ IE in (KJ/mol) is $A_2$

$A^{+2}(g) \longrightarrow A^{+3}(g) + e^{-}$ IE in (KJ/mol) is $A_3$

then correct order of IE is :-

(1) $A_1 > A_2 > A_3$ (2) $A_1 = A_2 = A_3$

(3) $A_1 < A_2 < A_3$ (4) $A_3 = A_2 < A_1$

**PT0108**

**92.** $IE_1$ $IE_2$ and $IE_3$ of an element are 10 eV, 15 eV, 45 eV respectively, the most stable oxidation state of the element will be :-

(1) + 1 (2) + 2 (3) + 3 (4) + 4

**PT0109**

**93.** Select the correct order of I.E. :-

(1) $Cl^{-} > Cl > Cl^{+}$ (2) $Cl^{+} > Cl > Cl^{-}$

(3) $Cl > Cl^{+} > Cl^{-}$ (4) $Cl^{-} > Cl^{+} > Cl$

**PT0110**

**94.** Incorrect statement is :

(1) The equation of 2nd IP is $X(g) \rightarrow X^{+2} + 2e^{-}$

(2) 2nd IP of an element lower than its 3rd IP

(3) Ionisation energy is expressed in units of kJ/mole

(4) Ionisation enthalpies are always positive for a neutral atom.

**PT0250**

**95.** In the curve of IP vs atomic no. (Z) :

(1) maxima would be for halogens.

(2) maxima would be for inert gases.

(3) minima would be for alkali metals.

(4) (2) and (3) both

**PT0251**

**96.** Choose correct statement :

(i) IP does not depend upon electron - electron repulsion.

(ii) IP is affected by the attraction of electron towards nucleus.

(iii) The $Z_{eff}$ will always less than nuclear charge for much electron species.

(iv) Generally, shielding is much more effective when orbitals in inner shell are partially filled.

(1) i, ii (2) ii, iii

(3) ii, iii, iv (4) ii, iv

**PT0252**

**97.** IP of B is slightly less than Be because :

(1) B has less $Z_{eff}$ than Be

(2) B has more size than that of Be

(3) 2p-electron of boron is more shielded from the nucleus by inner core of electron than the 2s-electrons of Be.

(4) Be has more stable configuration than B so it has more IP.

**PT0253**

**98.** The IP of oxygen is less than nitrogen. Reason is :

(1) elctronegativity of oxygen is more than nitrogen

(2) It is easier to remove the fourth 2p electron from oxygen due to increased electron-electron repulsion.

(3) $Z_{eff}$ of nitrogen is more

(4) Oxygen has more stable electronic configuration

**PT0254**

## ELECTRON AFFINITY

**99.** In the process $Cl_{(g)} + e^{-} \xrightarrow{\Delta H} Cl^{-}(g)$, $\Delta H$ is

(1) Positive (2) Negative

(3) Zero (4) None

**PT0111**

**100.** Process in which maximum energy is released:-

(1) $O \rightarrow O^{-2}$ (2) $Mg^{+} \rightarrow Mg^{+2}$

(3) $Cl \rightarrow Cl^{-}$ (4) $F \rightarrow F^{-}$

**PT0112**

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**101.** Which of the following is energy releasing process
(1) $X^- \rightarrow X(g) + e^-$ (2) $O^-(g) + e^- \rightarrow O^{2-}$
(3) $O(g) \rightarrow O^+(g) + e^-$ (4) $O(g) + e^- \rightarrow O^-(g)$
**PT0113**

**102.** In which of the following process energy is liberated:-
(1) $Cl \rightarrow Cl^+ + e$ (2) $HCl \rightarrow H^+ + Cl^-$
(3) $Cl + e \rightarrow Cl^-$ (4) $O^- + e \rightarrow O^{-2}$
**PT0114**

**103.** Element of which atomic number has highest electron affinity:-
(1) 35 (2) 17 (3) 9 (4) 53
**PT0115**

**104.** The electron affinity
(1) Of carbon is greater than oxygen
(2) Of fluorine is less than iodine
(3) Of fluorine is less than chlorine
(4) Of sulphur is less than oxygen
**PT0116**

**105.** Energy absorbed in second electron addition in an atom is called.
(1) 1st IP (2) 2nd EA
(3) 1st EA (4) 2nd IP
**PT0118**

**106.** Which of the following electronic configuration is expected to have highest electron affinity:-
(1) $2s^2\ 2p^0$ (2) $2s^2\ 2p^2$
(3) $2s^2\ 2p^3$ (4) $2s^2\ 2p^1$
**PT0120**

**107.** In which of the following process, least energy is required :-
(1) $F^-_{(g)} \longrightarrow F_{(g)} + e^-$ (2) $P^-_{(g)} \longrightarrow P_{(g)} + e^-$
(3) $S^-_{(g)} \longrightarrow S_{(g)} + e^-$ (4) $Cl^-_{(g)} \longrightarrow Cl_{(g)} + e^-$
**PT0122**

**108.** Which of the following pairs will have the most and least electron affinity respectively
(1) F, Cl (2) Cl, F (3) Cl, S (4) Cl, P
**PT0255**

**109.** In Group 16 elements, minimum $\Delta H_{eg}$ observed (with negative sign) for -
(1) Oxygen (2) Sulphur
(3) Selenium (4) Tellurium
**PT0256**

**110.** Select the correct statement regarding electron affinity-
(1) Electron affinity of F atom is less than that of Cl atom
(2) In 3rd period, electron-electron repulsion is much less than 2nd period
(3) Electron affinity depends on stability of configuration
(4) All of these
**PT0257**

**111.** (i) Process of adding electron to an atom may be exothermic or endothermic.
(ii) In most of the elements energy is released when an electron is added to the atom.
(iii) Halogens have maximum $\Delta H_{eg}$ in their respective period because of their stable electronic configuration.
(iv) Electron affinity order S < O
(v) $\Delta H_{eg}$ is maximum positive for inert gases in their respective period
Select the correct code in terms of true (T) and false (F)
(1) T F F T F (2) F T T F T
(3) T T F F T (4) F F F T F
**PT0258**

**112.** The electron gain enthalpy of S or Cl is more negative than that of preceding element because:
(1) the added electron goes to the smaller n = 2 and suffers significant repulsion from other electron present in this level.
(2) The added electron goes to n = 3 and occupies a larger region of space and electron-electron repulsion is much less.
(3) They show higher $Z_{eff}$
(4) They are having greater atomic radii.
**PT0259**

## ELECTRONEGATIVITY

**113.** The correct set of decreasing order of electronegativity is :-
(1) Li, H, Na (2) Na, H, Li
(3) H, Li, Na (4) Li, Na, H
**PT0123**

**114.** Polarity of a bond can be explained by :–
(1) Electron affinity (2) Ionisation potential
(3) Electronegativity (4) All of the above
**PT0124**

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**115**. Electronegativity values for elements are useful in predicting :-
(1) Bond energy of a molecule
(2) Polarity of a bond
(3) Nature of an oxide
(4) All

**PT0125**

**116**. Mulliken scale of electronegativity uses the concept of :-
(1) E. A. and EN of pauling
(2) E. A. and atomic size
(3) E.A. and I.P.
(4) E.A. and bond energy

**PT0126**

**117.** The pair with minimum difference in electronegativity is :-
(1) F, Cl (2) C,H (3) P, H (4) Na, Cs

**PT0127**

**118**. Least electronegative element is :-
(1) I (2) Br (3) C (4) Cs

**PT0128**

**119**. The nomenclature of ICl is iodine chloride because
(1) Size of I < Size of Cl
(2) Atomic number of I > Atomic number of Cl
(3) E.N. of I < E.N. of Cl
(4) E. A. of I < E. A. of Cl

**PT0130**

**120**. Among the following least and most polar bonds are respectively :-
(a) C – I (b) N – O
(c) C – F (d) P – F
(1) d and c (2) a and d
(3) b and d (4) b and c

**PT0131**

**121.** Electronegativity of an element can be measured using :-
(1) Pauling's scale (2) Mulliken's scale
(3) Both (4) None

**PT0134**

**122.** As we proceed across the period in periodic table, we find there is a decrease in :-
(1) Ionisation energy
(2) Electron affinity
(3) Electronegativity
(4) Atomic radii

**PT0135**

**123.** The electronegativities of the following elements: H, O, F, S and Cl increase in the order :-
(1) H < O < F < S < Cl
(2) Cl < H < O < F < S
(3) H < S < O < Cl < F
(4) H < S < Cl < O < F

**PT0137**

**124.** Which of the following is different from other three oxides :-
(1) MgO (2) SnO
(3) PbO (4) ZnO

**PT0138**

**125.** Out of C, Si, Ge, Sn and Pb are metallic
(1) Ge, Sn, Pb (2) Sn, Pb
(3) Ge, Pb (4) Ge, Sn

**PT0260**

**126.** The correct increasing order of metallic character of Si, Be, Mg, Na, P is
(1) P < Si < Be < Mg < Na
(2) Si < P < Be < Na < Mg
(3) Na < Mg < Be < Si < P
(4) Mg < Na < P < Si < Be

**PT0261**

**127.** The radius of which ion is closest to $Li^+$ ion ?
(1) $Na^+$ (2) $Be^{2+}$ (3) $Mg^{2+}$ (4) $Al^{+3}$

**PT0262**

**128.** The correct order of Electronegativity of underlined element.
(1) $\underline{C}_2H_4 < \underline{C}_2H_2$ (2) $\underline{O}F_2 < Na_2\underline{O}$
(3) $\underline{H}Cl < \underline{H}_2$ (4) $\underline{H}_2 < Na\underline{H}$

**PT0263**

**129.** Which of the following have no acidic or basic properties :-
(1) CO (2) $N_2O$ (3) NO (4) All

**PT0264**

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**130.** Which of the following does not act as an amphoteric oxide ?

(1) $Al_2O_3$ (2) ZnO (3) BeO (4) NO

**PT0265**

**131.** The anomalous behaviour of 2nd period elements from rest of the members of their respective families is due to :-

(1) small size

(2) large charge/radius ratio

(3) higher $z_{eff}$

(4) All

**PT0266**

**132.** Select the correct statement :-

(1) $Cl_2O_7$ is the most acidic oxide

(2) Non-metallic character decreases in a period

(3) BeO is a basic oxide

(4) All are correct

**PT0267**

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## EXERCISE-I (Conceptual Questions) ANSWER KEY

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | **4** | **3** | **1** | **1** | **2** | **2** | **4** | **2** | **4** | **3** | **2** | **3** | **2** | **1** | **2** |
| Que. | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Ans. | **1** | **3** | **2** | **3** | **1** | **3** | **1** | **3** | **1** | **2** | **3** | **3** | **2** | **2** | **1** |
| Que. | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 |
| Ans. | **3** | **2** | **2** | **3** | **2** | **4** | **2** | **2** | **1** | **4** | **1** | **3** | **3** | **1** | **2** |
| Que. | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| Ans. | **3** | **4** | **1** | **1** | **1** | **2** | **4** | **4** | **4** | **4** | **3** | **1** | **3** | **1** | **3** |
| Que. | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 |
| Ans. | **2** | **4** | **3** | **1** | **3** | **3** | **4** | **1** | **3** | **4** | **3** | **2** | **4** | **2** | **2** |
| Que. | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| Ans. | **4** | **3** | **2** | **4** | **1** | **3** | **3** | **2** | **3** | **1** | **2** | **1** | **1** | **3** | **3** |
| Que. | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 | 101 | 102 | 103 | 104 | 105 |
| Ans. | **3** | **2** | **2** | **1** | **4** | **2** | **3** | **2** | **2** | **3** | **4** | **3** | **2** | **3** | **2** |
| Que. | 106 | 107 | 108 | 109 | 110 | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 |
| Ans. | **2** | **2** | **4** | **1** | **4** | **3** | **2** | **3** | **3** | **4** | **3** | **3** | **4** | **3** | **2** |
| Que. | 121 | 122 | 123 | 124 | 125 | 126 | 127 | 128 | 129 | 130 | 131 | 132 | | | |
| Ans. | **3** | **4** | **4** | **1** | **2** | **1** | **3** | **1** | **4** | **4** | **4** | **1** | | | |

## EXERCISE-III (Analytical Questions) Master Your Understanding

**1.** Which of the following is correct match :-

| | Atomic number | Group number | Period number |
|---|---|---|---|
| (A) | 46 | 10 | 6 |
| (B) | 58 | 3 | 6 |
| (C) | 56 | 2 | 6 |
| (D) | 42 | 6 | 5 |

(1) Only B, C, D (2) Only A, B, C
(3) Only B, C (4) Only A, C, D

**PT0172**

**2.** Atomic number of Ag is 47. In the same group the atomic number of elements placed above and below Ag will be :-
(1) 37, 67 (2 ) 29, 79 (3) 39, 69 (4) 29, 65

**PT0174**

**3.** Which of the following statements is wrong :-
(1) van der Waal's radius of iodine is more than its covalent radius
(2) All isoelectronic ions belong to same period of the periodic table
(3) $IE_1$ of N is higher than that of O while $IE_2$ of O is higher than that of N
(4) The electron affinity of N is less than that of P

**PT0178**

**4.** The inter nuclear distance in $H_2$ and $Cl_2$ molecules are 74 and 198 Å. respectively. The bond length of HCl may be
(EN of H = 2.1 Cl = 3.0)
(1) 136 Å (2) 272 Å
(3) 135.919 Å (4) 271.919 Å

**PT0179**

**5.** These are 3 elements A, B and C. Their atomic number are $Z_1$, $Z_2$, $Z_3$ respectively. If $Z_3 - Z_1 = 2$ and $\frac{Z_1 + Z_3}{2} = Z_2$ and the electronic configuration of element C is [Ar] $3d^2 4s^2$ then correct order of atomic radius is :-
(1) $A^{+2} < B^{+3} < C^{+4}$ (2) $A^{+2} = B^{+3} = C^{+4}$
(3) $A^{+2} > B^{+3} > C^{+4}$ (4) $B^{+3} < A^{+2} = C^{+4}$

**PT0180**

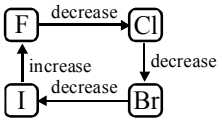
**6.** $M(g) \rightarrow M^+(g) + e^-$, $\Delta H = 100$ eV
$M(g) \rightarrow M^{2+}(g) + 2e^-$, $\Delta H = 250$ eV
Which is incorrect statements :-
(1) $I_1$ of M(g) is 100 eV
(2) $I_2$ of M(g) is 150 eV
(3) $I_2$ of M(g) is 250 eV
(4) none

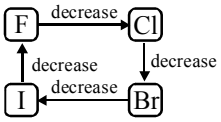
**PT0181**

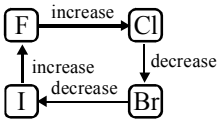
**7.** Successive ionisation energies of an element A are 100 eV, 150 eV, 181 eV, 2000 eV, 2200 eV correct statement regarding A is :-
(1) Element 'A' may be metal
(2) Formula of oxide of A may be $A_2O_3$
(3) Oxide of element A may be amphoteric
(4) All are correct

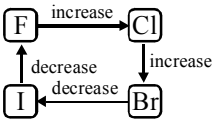
**PT0182**

**8.** Which of the following diagrams is correctly related to electron affinity of halogens :-

(1) 


(2) 


(3) 


(4) 


**PT0184**

**9.** Elements of which group form anions most readily:-
(1) Oxygen family (2) Nitrogen group
(3) Halogens (4) Alkali metals

**PT0185**

**10.** Which is the weakest base among NaOH, $Ca(OH)_2$, KOH and $Zn(OH)_2$:-
(1) NaOH (2) KOH
(3) $Ca(OH)_2$ (4) $Zn(OH)_2$

**PT0186**

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**11.** Identify the incorrect are :-
(1) Shielding constant ($\sigma$) : Li < Na < K < Rb
(2) $Z_{eff}$ : Li > Na > K > Rb
(3) Ionic radius : $O^{2-} > F^- > Na^+ > Mg^{2+}$
(4) Atomic size : Li < Na < K < Rb

**PT0188**

**12.** If electronegativity values of element X and Y are 3.8 and 1.8 respectively, then percentage of ionic character in compound XY is :
(1) 50 (2) 46 (3) 64 (4) 36

**PT0189**

**13.** The order of ionisation potential between $He^+$ ion and H-atom (both species are in gaseous state) is:-
(1) I.P. ($He^+$) = I.P. (H) (2) I.P. ($He^+$) < I.P. (H)
(3) I.P. ($He^+$) > I.P. (H) (4) Cannot be compared

**PT0191**

**14.** Electronic configuration are :-
A - $1s^2 2s^2 2p^1$
B - $1s^2 2s^2 2p^6 3s^1 3p^2$
C - $1s^2 2s^1 2p^1$
D - $1s^2 2s^2 2p^5 3s^1$
then which among these will belong to the same group in the periodic table ?
(1) A & B (2) A, B, C
(3) A, B, D (4) A, B, C, D

**PT0192**

**15.** The $IE_1$ & $IE_2$ of three elements A, B & C are given as (IE in KJ/mol).

| | A | B | C |
|---|---|---|---|
| $IE_1$ | 400 | 550 | 1150 |
| $IE_2$ | 2650 | 1070 | 2090 |

Identify the element which represent a non-metal:-
(1) A (2) B
(3) Both A & B (4) C

**PT0193**

**16.** In which of the following arrangements the order is NOT according to the property indicated against it?
(1) $Al^{3+} < Mg^{2+} < Na^+ < F^-$ - increasing ionic size
(2) B < C < N < O - increasing first ionization energy
(3) I < Br < F < Cl - increasing electron gain ethalpy (with negative sign)
(4) Li < Na < K < Rb - increasing metallic radius

**PT0195**

**17.** Which one of the following orders represents the correct sequence of the increasing basic nature of the given oxides ?
(1) $Na_2O < K_2O < MgO < Al_2O_3$
(2) $K_2O < Na_2O < Al_2O_3 < MgO$
(3) $Al_2O_3 < MgO < Na_2O < K_2O$
(4) $MgO < K_2O < Al_2O_3 < Na_2O$

**PT0196**

**18.** The outer electron configuration of Gd (Atomic No. : 64) is :-
(1) $4f^4\ 5d^4\ 6s^2$ (2) $4f^7\ 5d^1\ 6s^2$
(3) $4f^3\ 5d^5\ 6s^2$ (4) $4f^8\ 5d^0\ 6s^2$

**PT0197**

**19.** The correct order of electron gain enthalpy with negative sign of F, Cl, Br and I, having atomic number 9, 17, 35 and 53 respectively, is :-
(1) I > Br > Cl > F (2) F > Cl > Br > I
(3) Cl > F > Br > I (4) Br > Cl > I > F

**PT0198**

**20.** The first ionisation potential of Na is 5.1 eV. The value of electron gain enthalpy of $Na^+$ will be :-
(1) – 2.55 eV (2) – 5.1 eV
(3) – 10.2 eV (4) + 2.55 eV

**PT0199**

**21.** For electron affinity of halogens which of the following is correct :-
(1) Br > F (2) F > Cl
(3) Br > Cl (4) F > I

**PT0200**

**22.** The pair of amphoteric hydroxide is
(1) $Al(OH)_3$ , LiOH
(2) $Be(OH)_2$ , $Mg(OH)_2$
(3) $B(OH)_3$ , $Be(OH)_2$
(4) $Be(OH)_2$ , $Zn(OH)_2$

**PT0201**

**23.** Electronegativity is the measurement of capacity of an atom by which :
(1) Electrons get repelled
(2) Electrons get attracted
(3) Gain of electron
(4) Loose of proton

**PT0202**

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**24.** The ions $O^{2-}$, $F^{-}$, $Na^{+}$, $Mg^{2+}$ and $Al^{3+}$ are isoelectronic. Their ionic radii show

(1) an increase from $O^{2-}$ to $F^{-}$ and then decrease from $Na^{+}$ to $Al^{3+}$

(2) a decrease from $O^{2-}$ to $F^{-}$ and then increase from $Na^{+}$ to $Al^{3+}$

(3) a significant increase from $O^{2-}$ to $Al^{3+}$

(4) a significant decrease from $O^{2-}$ to $Al^{3+}$

**PT0203**

**25.** Ionic radii are :-

(1) Directly proportional to square of effective nuclear charge

(2) Inversely proportional to effective nuclear charge

(3) Inversely proportional to square of effective nuclear charge

(4) Directly proportional to effective nuclear charge

**PT0204**

**26.** Which one of the following arrangements represents the correct order of electron gain enthalpy (with negative sign) of the given atomic species :-

(1) F < Cl < O < S (2) S < O < Cl < F

(3) O < S < F < Cl (4) Cl < F < S < O

**PT0205**

**27.** Which of the following E.A. order is not correct ?

(1) N < O < S (2) Cl > O > N > C

(3) O < S < F < Cl (4) B < C < Si < S

**PT0209**

**28.** Which one of the following order of given properties is correct ?

(1) Atomic radius - Li < Be < B

(2) Ionisation potential - Li < Be < B

(3) Electron affinity -Li < Be < B

(4) Electronegativity - Li < Be < B

**PT0210**

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## EXERCISE-III — ANSWER KEY

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | **1** | **2** | **2** | **3** | **3** | **3** | **4** | **3** | **3** | **4** | **2** | **2** | **3** | **1** | **4** |
| Que. | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | | |
| Ans. | **2** | **3** | **2** | **3** | **2** | **4** | **4** | **2** | **4** | **2** | **3** | **2** | **4** | | |